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1. if we insert an electrode into an egg and place a second electrode outside it, we can measure the constant difference in charge across the egg _____**Plasma Membrane**
2. Bleil and Wassarman (_____) isolated an 83-kDa glycoprotein, ZP3, from the mouse zona that was the active site for sperm binding.The other two zona glycoproteins they found, ZP1 and ZP2, failed to compete for sperm binding?_____ **1988**
3. A third protein released by the cortical granules, a _____, hardens the fertilization envelope by crosslinking tyrosine residues on adjacent proteins?_____ **Peroxidase enzyme**
4. The sea urchin egg has ___ mechanisms to avoid polyspermy?_____**2**
5. In capacitation, the _____, protein phosphorylation occurs?_____ **Fourth**
6. The entrance of multiple sperm polyspermy leads to disastrous consequences in most organisms. In the sea urchin, fertilization by two sperm results in a _____ nucleus, in which each chromosome is represented three times rather than twicen_____**Triploid**
7. The fast sperm inhibition condition is maintained by the plasma membrane. which steadfastly inhibits the entry of …..ions into the oocyte and prevents potassium ions from leaking out into the environment. _____**Sodium**
8. The difference between the acrosomal reaction in sea urchins and mammals may be due to the thickness of the extracellular envelopes surrounding the egg. In the sea urchin, the vitelline envelope is very thin and porous?_____ **True**
9. In capacitation, the _____, the membrane potential of the sperm becomes more negative as potassium ions leave the sperm?_____ **Third**
10. The sperm and egg plasma membranes then join together, and material from the sperm membrane can later be found on the egg membrane. The sperm nucleus and tail pass through the resulting cytoplasmic bridge, which is widened by the _____?____ **Actin polymerization**
11. Directly beneath the sea urchin egg plasma membrane are about …cortical granules____**15,000.**
12. The ____ is the specific glycoprotein in the mouse zona pellucida to which the sperm bind._____ **ZP3**
13. The Fast block to polyspermy is achieved by changing the electric potential of the egg plasma membrane. This membrane provides a selective barrier between the egg cytoplasm and the outside environment, and the ionic concentration of the egg differs greatly from that of its surroundings?_____ **True**
14. The egg's microvilli attach to……..provides support for the blastomeres _____**hyaline layer.**
15. in the sea urchin, all regions of the egg plasma membrane are capable of fusing with sperm. In several other species, certain regions of the membrane are specialized for sperm recognition and fusion. Fusion is an active process, often mediated by specific "fusogenic proteins? _____ _____**True**
16. The Ferilin _____ the sperm to the egg by binding to the 6β1 integrin protein on the egg plasma membrane. Moreover, like sea urchin bindin (to which it is not structurally related), fertilin has a hydrophobic region that could potentially mediate the union of the two membranes._____ **Adheres**
17. The removal of sperm is accomplished by the cortical granule reaction, a slower, mechanical block to _____ that becomes active about a minute after the first successful sperm-egg attachment._____ **Polyspermy**
18. Recognition of sperm by the vitelline envelope or zona pellucida is followed by the lysis of that portion of the envelope or zona in the region of the sperm head by the enzymes? ____**Acrosomal**
19. In capacitation, the _____, particular proteins or carbohydrates on the sperm surface are lost during capacitation?_____ **Second**
20. release of calcium from intracellular storage can be monitored visually using calcium-activated luminescent dyes such as_____**fura-2**
21. The drug is .. calcium ionophore (a compound that transports free calcium ions across lipid membranes)_____**A23187**
22. The _____ is a glycoprotein matrix, which is synthesized and secreted by the growing oocyte?_____ **Zona pellucida**

23. The Wilcox and colleagues…. found that nearly all human pregnancies result from sexual intercourse during a 6-day period ending on the day of ovulation?_____**1995**
24. Finally, a fourth cortical granule protein, _____, forms a coating around the egg?_____ **Hyalin**
25. The molecular changes that account for capacitation are of _____types?_____**4**
26. It appears that the acrosome-reacted sperm transfer their binding from ZP3 to the adjacent _____ molecules. After a mouse sperm has entered the egg, the egg cortical granules release their contents._____ **ZP2**
27. The set of physiological changes that allow the sperm to be competent to fertilize the egg is called capacitation. The requirement for capacitation varies from species to species?_____ **True**
28. Using immunological techniques,….demonstrated that bindin is located specifically on the acrosomal process exactly where it should be for sperm-egg recagnition?____**Moy and Vacquier**
29. in capacitation, the ……. the fluidity of the sperm plasma membrane is altered by the removal of cholesterol by albumin proteins found in the female reproductive tract? _____**First**
30. In mammals, the _____ proteins in the sperm plasma membrane are essential for sperm membrane-egg membrane fusion?_____ **Fertilin**
31. The _____ released by the cortical granules produce an osmotic gradient that causes water to rush into the space between the plasma membrane and the vitelline envelope, causing the envelope to expand and become the fertilization envelope._____ **Mucopolysaccharides**
32. The Fast block to polyspermy is achieved by changing the electric potential af the egg plasma membrane. This membrane provides a selective barrier between the egg cytoplasm and the outside environment, and the ionic concentration of the eqg differs greatly from that of its surroundings? _____**True**